

Component-wise Approximate Bayesian Computation via Gibbs-like steps

Comparative study of ABC-Gibbs versus SMC-ABC on hierarchical models

Project 4 — *Gibbs and ABC*
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Abstract

This project provides a rigorous comparative study of the ABC-GIBBS algorithm introduced by Clarté *et al.* (2021) against an adaptive SMC-ABC sampler (Del Moral *et al.*, 2012; Toni *et al.*, 2008), on two hierarchical models. The first is the hierarchical Gaussian model from Section 3.2 of the reference paper, whose posterior is analytically known, allowing for the quantification of the *exact* inferential error. The second is the simple hierarchical *g*-and-*k* model, which is intractable, where we use the predictive *a posteriori* distance as a goodness-of-fit indicator. The comparison is based on three metrics: (i) computational cost (CPU time and number of model simulations), (ii) inferential error (bias, variance ratio, Wasserstein-1 distance to the target distribution), and (iii) Monte Carlo error, estimated via the batch means method (Flegal & Jones, 2010). On the Gaussian model in dimension $1 + n = 21$, ABC-Gibbs vastly outperforms SMC-ABC at a comparable cost: the average Wasserstein distance on the μ_j 's is 0.024 versus 0.922, for a simulation cost that is more than a factor of 2 *lower*. On the hierarchical *g*-and-*k* model, the average predictive distance is 352 versus 1082, confirming the progressive collapse of SMC-ABC in increasing dimensions. These results quantitatively reproduce the observations made by Clarté *et al.* (2021) and illustrate the relevance of a component-by-component strategy to break the curse of dimensionality in likelihood-free inference.

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1 Introduction and Problem Statement

1.1 General Framework of Likelihood-Free Inference

Classical Bayesian inference relies on the evaluation of the posterior

$$\pi(\theta \mid x^*) \propto \pi(\theta) f(x^* \mid \theta),$$

where π denotes the *prior distribution* on the parameter $\theta \in \Theta$ and $f(\cdot \mid \theta)$ the likelihood of the observed data x^* . In many modern models — generative models in population genetics (Tavaré *et al.*, 1997; Beaumont *et al.*, 2002), biological system dynamics (Toni *et al.*, 2008), epidemiological models, g -and- k distributions (Prangle, 2017) — the likelihood cannot be computed in a reasonable amount of time, but we have access to a *simulator*: given θ , we can produce a realization $x \sim f(\cdot \mid \theta)$. This is exactly the context of likelihood-free inference, which APPROXIMATE BAYESIAN COMPUTATION (ABC) aims to address.

The core idea, pioneered by Tavaré *et al.* (1997) and later Beaumont *et al.* (2002), is disarmingly simple: draw a large number of pairs $(\theta^{(i)}, x^{(i)})$ from the prior predictive $\pi(\theta)f(x \mid \theta)$, and retain only the parameters $\theta^{(i)}$ for which the pseudo-data $x^{(i)}$ falls within a neighborhood of the observations x^* , most often via a summary statistic $s(\cdot)$ and a tolerance $\varepsilon > 0$.

1.2 The Curse of Dimensionality in ABC

The major drawback of *vanilla* ABC methods is their dramatic sensitivity to the dimension of the parameter. Fearnhead & Prangle (2012) and Li & Fearnhead (2018) show that, for an optimal approximation, the dimension of the summary s must *coincide* with that of θ . However, for a fixed summary dimension, the probability that a simulation falls within a Euclidean ball of radius ε around $s(x^*)$ decreases exponentially with this dimension, leading to a prohibitive computational cost.

Clarté *et al.* (2021) proposed an elegant strategy to break this curse: exploiting the conditional structure of the joint posterior by performing Gibbs-like steps, each updating a single component θ_j based on a summary statistic s_j of *small* dimension. This approach, dubbed ABC-GIBBS, sacrifices some of the classical theoretical guarantees of the Gibbs sampler (possible incompatibility of the conditionals) in favor of substantial numerical efficiency. The authors nonetheless establish the total variation convergence of the chain under reasonable assumptions, and empirically demonstrate the method’s superiority over SMC-ABC in several hierarchical models.

1.3 Purpose of the Project

In accordance with the instructions for *Project 4 — Gibbs and ABC* from the Simulation and Monte Carlo course, we aim to compare three algorithms:

1. the ABC-GIBBS algorithm by Clarté *et al.* (2021) (reference algorithm);
2. an ABC sampler more sophisticated than vanilla rejection: we chose adaptive SMC-ABC in the sense of Del Moral *et al.* (2012); Toni *et al.* (2008), which represents the state of the art in sequential ABC and serves as the main competitor in the reference paper;
3. an *exact gold standard*. The prompt suggests a Random-Walk Metropolis, justified by the fact that the likelihood is tractable in the toy model. We go one step further: for the hierarchical Gaussian model in Section 3.2, the posterior is conjugate and *analytically available*. We therefore use it directly as a baseline, which eliminates any calibration or mixing errors from a Metropolis competitor; for the g -and- k model, however, no analytical reference exists, and we revert to the predictive metric used by Clarté *et al.* (2021).

Three metrics are systematically reported, as requested: (i) CPU time and number of model simulations, (ii) inferential error (bias, variance ratio, Wasserstein-1 distance), (iii) intra-chain

Monte Carlo error via batch means. An independent replicate study empirically confirms the relevance of the Monte Carlo error estimates. Finally, as a bonus, the simple hierarchical g -and- k model — featured in Section 4 of the paper — is treated, illustrating the robustness of the conclusion on a truly intractable model.

2 Theoretical Framework: ABC, Gibbs, ABC-Gibbs

2.1 Vanilla ABC

Definition 2.1. ABC Distribution

Given a summary statistic $s : \mathcal{X} \rightarrow \mathbb{R}^q$, a distance d on $\mathbb{R}^q$, and a tolerance $\varepsilon > 0$, the *ABC distribution* is defined as the distribution on Θ with density

$$\pi_\varepsilon(\theta \mid s(x^*)) \propto \pi(\theta) \int_{\mathcal{X}} f(x \mid \theta) \mathbb{1}_{\{d(s(x), s(x^*)) < \varepsilon\}} dx.$$

Proposition 2.2. Limit of ABC methods

Under standard regularity assumptions (Frazier *et al.*, 2018), as $\varepsilon \downarrow 0$, $\pi_\varepsilon(\cdot \mid s(x^*))$ converges in total variation to the posterior $\pi(\cdot \mid s(x^*))$, which coincides with the exact posterior $\pi(\cdot \mid x^*)$ if and only if s is sufficient for θ .

Algorithm 1: Rejection ABC (vanilla)

Input: observations x^* , size N , tolerance $\varepsilon > 0$, statistic s , distance d .

Output: sample $(\theta^{(1)}, \dots, \theta^{(N)})$.

```

1 for  $i = 1, \dots, N$  do
2   repeat
3     Draw  $\theta^{(i)} \sim \pi(\cdot)$ ;
4     Draw  $x^{(i)} \sim f(\cdot \mid \theta^{(i)})$ ;
5   until  $d(s(x^{(i)}), s(x^*)) < \varepsilon$ ;
6 end
```

In practice, since the threshold ε can rarely be set *a priori*, we proceed via a *reference table*: we draw a large number of pairs $(\theta^{(i)}, x^{(i)})$ and keep those for which the distance $d(s(x^{(i)}), s(x^*))$ falls in the lower empirical quantile $\alpha \in (0, 1)$ of the sample. This procedure is equivalent to setting ε as an empirical quantile of the predictive distribution.

2.2 Gibbs Sampler

Definition 2.3. Gibbs Sampler

Let $\theta = (\theta_1, \dots, \theta_n)$ and π be a target distribution on $\Theta = \prod_{j=1}^n \Theta_j$ admitting simulatable full conditionals $\pi(\theta_j \mid \theta_{-j})$. The *systematic Gibbs sampler* produces a Markov chain $(\theta^{(t)})_{t \geq 0}$ via cyclical updates:

$$\theta_j^{(t)} \sim \pi\left(\cdot \mid \theta_1^{(t)}, \dots, \theta_{j-1}^{(t)}, \theta_{j+1}^{(t-1)}, \dots, \theta_n^{(t-1)}\right), \quad j = 1, \dots, n.$$

The major appeal of the Gibbs sampler lies in its adaptation to hierarchical structures: if we consider the model from [Clarté et al. \(2021\)](#), namely

$$x_j \mid \mu_j \sim f(\cdot \mid \mu_j), \quad \mu_j \mid \alpha \stackrel{\text{iid}}{\sim} \pi(\cdot \mid \alpha), \quad \alpha \sim \pi(\alpha), \quad (\text{H})$$

the posterior of $\mu = (\mu_1, \dots, \mu_n)$ conditioned on α factorizes:

$$\pi(\mu \mid x_1, \dots, x_n, \alpha) \propto \prod_{j=1}^n \pi(\mu_j \mid \alpha) f(x_j \mid \mu_j).$$

We recognize a decomposition into n *independent* 1-dimensional problems — this is exactly the property that makes Gibbs so efficient in hierarchical settings. When neither $\pi(\mu_j \mid \alpha)f(x_j \mid \mu_j)$ nor $\pi(\alpha \mid \mu)$ are available in closed form — as is the case when the likelihood f is intractable —, the classical Gibbs sampler cannot be used. This is precisely the bottleneck that ABC-Gibbs removes.

2.3 ABC-Gibbs

Algorithm 2: ABC-Gibbs ([Clarté et al., 2021](#))

Input: x^* , size N , initial point $\theta^{(0)}$, thresholds $\varepsilon_1, \dots, \varepsilon_n$, summaries $s_1, \dots, s_n$, distances $d_1, \dots, d_n$.
Output: $(\theta^{(1)}, \dots, \theta^{(N)})$.

```

1 for  $i = 1, \dots, N$  do
2   for  $j = 1, \dots, n$  do
3     Draw  $\theta_j^{(i)} \sim \pi_{\varepsilon_j} \left\{ \cdot \mid s_j \left( x^*, \theta_1^{(i)}, \dots, \theta_{j-1}^{(i)}, \theta_{j+1}^{(i-1)}, \dots, \theta_n^{(i-1)} \right) \right\}$ ;
4   end
5 end
```

The algorithm replaces each call to the exact full conditional with a call to its ABC version: we draw a mini-reference table from the *conditional* predictive (rather than the *prior* predictive, as in vanilla ABC), and then retain the argument that minimizes the corresponding summary distance. The key to the method is threefold:

- Since each θ_j is typically 1-dimensional, the summary s_j can be too, per [Fearnhead & Prangle \(2012\)](#): the curse is contained;
- The conditional predictive is centered on the *current* values of the other components, making it far more efficient than the prior predictive;
- In the hierarchical case (H), some conditionals might be available in closed form, bypassing the need for an ABC step altogether.

2.4 Convergence Theorem (case $n = 2$)

The foundational result of the paper is as follows.

Theorem 2.4. Convergence of ABC-Gibbs, $n = 2$ (Th. 1 from [Clarté et al.](#))

Suppose there exists $0 < \kappa < 1/2$ such that

$$\sup_{\theta_1, \tilde{\theta}_1} \left\| \pi_{\varepsilon_2} \left\{ \cdot \mid s_2(x^*, \theta_1) \right\} - \pi_{\varepsilon_2} \left\{ \cdot \mid s_2(x^*, \tilde{\theta}_1) \right\} \right\|_{\text{TV}} = \kappa.$$

Then the Markov chain produced by Algorithm 2 converges geometrically in total variation to a stationary distribution ν_ε , with rate $1 - 2\kappa$.

Proof. We outline the proof, omitting the conditioning on x^*, s_1, s_2 to lighten the notation. Let Q be the transition kernel of the marginal chain on Θ_1 (we can reduce to this marginal since the mapping $\theta_1 \mapsto \theta_2 \mapsto \theta'_1$ entirely characterizes the dynamics). We show that $Q : \mathcal{P}(\Theta_1) \rightarrow \mathcal{P}(\Theta_1)$ is a contraction with constant $1 - 2\kappa$ under the total variation distance; Banach's fixed-point theorem then provides the existence and uniqueness of ν_ε as well as geometric convergence.

Construction of a coupling. For two measures $\nu, \tilde{\nu} \in \mathcal{P}(\Theta_1)$, let ξ_0 be the optimal coupling between ν and $\tilde{\nu}$, i.e., the distribution on $\Theta_1 \times \Theta_1$ achieving the lower bound $\mathbb{P}_{\xi_0}(\theta_1 \neq \tilde{\theta}_1) = 2\|\nu - \tilde{\nu}\|_{\text{TV}}$ (existence is guaranteed for Polish spaces).

We define the coupling kernel $\tilde{Q}$ via the following procedure:

1. Draw $(\theta_1, \tilde{\theta}_1) \sim \xi_0$;
2. Draw $(\theta_2, \tilde{\theta}_2) \sim \xi_1(\cdot \mid \theta_1, \tilde{\theta}_1)$ where ξ_1 is an *optimal* coupling between $\pi_{\varepsilon_2}(\cdot \mid \theta_1)$ and $\pi_{\varepsilon_2}(\cdot \mid \tilde{\theta}_1)$;
3. Draw $(\theta'_1, \tilde{\theta}'_1) \sim \xi_2(\cdot \mid \theta_2, \tilde{\theta}_2)$ where ξ_2 is an optimal coupling between $\pi_{\varepsilon_1}(\cdot \mid \theta_2)$ and $\pi_{\varepsilon_1}(\cdot \mid \tilde{\theta}_2)$.

This construction satisfies the *absorption* property: if $\theta_2 = \tilde{\theta}_2$, then $\theta'_1 = \tilde{\theta}'_1$ almost surely, because the optimal coupling between a distribution and itself is concentrated on the diagonal, $(x, y) \mapsto \nu_0(x) \delta_{y=x}$.

Upper bounding the decoupling probability. By the definition of total variation and the coupling nature of $\tilde{Q}\xi_0$:

$$\|Q\nu - Q\tilde{\nu}\|_{\text{TV}} \leq \frac{1}{2} \mathbb{P}_{\tilde{Q}\xi_0}(\theta'_1 \neq \tilde{\theta}'_1).$$

Decomposing based on whether the initial values were equal or not:

$$\begin{aligned} \mathbb{P}_{\tilde{Q}\xi_0}(\theta'_1 \neq \tilde{\theta}'_1) &= \underbrace{\mathbb{P}(\theta'_1 \neq \tilde{\theta}'_1 \mid \theta_1 = \tilde{\theta}_1)}_{=0 \text{ by absorption}} \mathbb{P}_{\xi_0}(\theta_1 = \tilde{\theta}_1) \\ &\quad + \mathbb{P}(\theta'_1 \neq \tilde{\theta}'_1 \mid \theta_1 \neq \tilde{\theta}_1) \mathbb{P}_{\xi_0}(\theta_1 \neq \tilde{\theta}_1). \end{aligned}$$

Since $\mathbb{P}_{\xi_0}(\theta_1 \neq \tilde{\theta}_1) = 2\|\nu - \tilde{\nu}\|_{\text{TV}}$, we get

$$\|Q\nu - Q\tilde{\nu}\|_{\text{TV}} \leq \|\nu - \tilde{\nu}\|_{\text{TV}} \mathbb{P}(\theta'_1 \neq \tilde{\theta}'_1 \mid \theta_1 \neq \tilde{\theta}_1).$$

Bound on the last probability. If coupling step 2 is successful, meaning $\theta_2 = \tilde{\theta}_2$, then by absorption in step 3, $\theta'_1 = \tilde{\theta}'_1$. The probability of overall failure is thus bounded by the probability of failure at step 2:

$$\mathbb{P}(\theta'_1 \neq \tilde{\theta}'_1 \mid \theta_1 \neq \tilde{\theta}_1) \leq \mathbb{P}(\theta_2 \neq \tilde{\theta}_2 \mid \theta_1 \neq \tilde{\theta}_1).$$

By the optimality of ξ_1 ,

$$\mathbb{P}(\theta_2 \neq \tilde{\theta}_2 \mid \theta_1, \tilde{\theta}_1) = 2 \|\pi_{\varepsilon_2}(\cdot \mid \theta_1) - \pi_{\varepsilon_2}(\cdot \mid \tilde{\theta}_1)\|_{\text{TV}} \leq 2\kappa,$$

by hypothesis. Integrating over $(\theta_1, \tilde{\theta}_1)$, we get $\mathbb{P}(\theta_2 \neq \tilde{\theta}_2 \mid \theta_1 \neq \tilde{\theta}_1) \leq 2\kappa$.

Conclusion. We obtain

$$\|Q\nu - Q\tilde{\nu}\|_{\text{TV}} \leq (1 - 2\kappa) \|\nu - \tilde{\nu}\|_{\text{TV}},$$

which represents the *minimum coupling probability*. Since $\kappa < 1/2$, $1 - 2\kappa \in (0, 1)$: Q is a strict contraction. Because $(\mathcal{P}(\Theta_1), \|\cdot\|_{\text{TV}})$ is complete, and the subset of probability measures is stable under Q and closed for $\|\cdot\|_{\text{TV}}$, Banach's theorem yields the existence of a unique fixed point $\nu_\varepsilon^{(1)}$. The theorem for the full chain on $\Theta_1 \times \Theta_2$ is immediately deduced. $\square$

2.5 Extension to the $n > 2$ case and limit bound

Theorem 2.5. Convergence in the general case (Th. 2)

Suppose that for all $\ell \leq n$,

$$\kappa_\ell = \sup_{\theta_{>\ell}, \tilde{\theta}_{>\ell}} \sup_{\theta_{<\ell}} \|\pi_{\varepsilon_\ell} \{\cdot \mid s_\ell(x^*, \theta_{<\ell}, \theta_{>\ell})\} - \pi_{\varepsilon_\ell} \{\cdot \mid s_\ell(x^*, \theta_{<\ell}, \tilde{\theta}_{>\ell})\}\|_{\text{TV}} < \frac{1}{2}.$$

Then the chain generated by Algorithm 2 converges geometrically in total variation to a stationary distribution ν_ε , with rate $1 - \prod_\ell 2\kappa_\ell$.

The coupling argument is analogous: we try to successively couple the components θ_ℓ , exploiting the fact that the already coupled components $\theta_{<\ell}$ have a chance of propagating the success. The condition for $\ell = n$ is trivially verified, which allows the recurrence to terminate properly.

Theorem 2.6. Distance between the limit distribution and the ABC posterior (Th. 3)

Suppose (for $n = 2$):

$$\begin{aligned} L_0 &= \sup_{\varepsilon_2} \sup_{\theta_1, \tilde{\theta}_1} \|\pi_{\varepsilon_2}(\cdot \mid s_2(x^*, \theta_1)) - \pi_0(\cdot \mid s_2(x^*, \tilde{\theta}_1))\|_{\text{TV}} < \frac{1}{2}; \\ L_1(\varepsilon_1) &= \sup_{\theta_2} \|\pi_{\varepsilon_1}(\cdot \mid s_1(x^*, \theta_2)) - \pi_0(\cdot \mid s_1(x^*, \theta_2))\|_{\text{TV}} \xrightarrow{\varepsilon_1 \rightarrow 0} 0; \\ L_2(\varepsilon_2) &= \sup_{\theta_1} \|\pi_{\varepsilon_2}(\cdot \mid s_2(x^*, \theta_1)) - \pi_0(\cdot \mid s_2(x^*, \theta_1))\|_{\text{TV}} \xrightarrow{\varepsilon_2 \rightarrow 0} 0. \end{aligned}$$

Then

$$\|\nu_\varepsilon - \nu_0\|_{\text{TV}} \leq \frac{L_1(\varepsilon_1) + L_2(\varepsilon_2)}{1 - 2L_0} \xrightarrow{\varepsilon \rightarrow 0} 0.$$

Proof idea. We resume the coupling strategy from Theorem 2.4 but this time coupling two different chains: one following the approximated kernel Q_ε , the other following the “exact” kernel Q_0 (the limit when all tolerances drop to 0). We show that $\mathbb{P}(\theta'_1 \neq \tilde{\theta}'_1 \mid \theta_1 = \tilde{\theta}_1) \leq 2L_1(\varepsilon_1) + 2L_2(\varepsilon_2)$ by bounding each step with the distances L_1, L_2 . The geometry of the contraction yields the factor $1/(1 - 2L_0)$. $\square$

2.6 Hierarchical Case: Theorem 4 and weakened conditions

For model (H) with a single layer, the conditional $\pi(\mu \mid x^*, \alpha)$ can be exactly simulated in many cases (notably the Gaussian-Gaussian model). A stronger result is then available.

Theorem 2.7. Hierarchical case with exact conditional (Th. 4)

Suppose there exists a non-empty convex set C with positive prior measure such that

$$\kappa_1 = \inf_{s_\alpha(\mu) \in C} \pi(B_{s_\alpha(\mu), \varepsilon_\alpha/4}) > 0, \quad \kappa_2 = \inf_{\alpha} \inf_{s_\alpha(\mu) \in C} \pi_{\varepsilon_\mu} \{B_{s_\alpha(\mu), 3\varepsilon_\alpha/2} \mid s_\mu(x^*, \alpha), \alpha\} > 0,$$

$$\kappa_3 = \inf_{\alpha} \pi_{\varepsilon_\mu} \{s_\alpha(\mu) \in C \mid s_\mu(x^*, \alpha)\} > 0,$$

where $B_{z,h}$ denotes the Euclidean ball centered at z with radius h . Then the chain from Algorithm 4 in the paper converges geometrically in total variation to a stationary distribution ν_ε , at a rate of $1 - \kappa_1 \kappa_2 \kappa_3^2$.

2.7 Pathology: an illuminating counter-example

The paper presents an instructive counter-example in Section 12 of the supplementary material: a single observation $x^* = 5$ from a mixture $\frac{1}{2}\mathcal{U}(\theta_1, \theta_1 + 1) + \frac{1}{2}\mathcal{U}(\theta_2, \theta_2 + 1)$, with a uniform prior over $A = \{(\theta_1, \theta_2) : 0 \leq \theta_1, \theta_2 \leq 10, |\theta_1 - \theta_2| > 2\}$. The exact posterior is uniform over the union of two symmetric bands. For $\varepsilon = 0.5$, vanilla ABC correctly captures both bands; ABC-Gibbs, however, captures only one (depending on random initialization), because the supports of the two conditionals can be disjoint, violating the assumption $\kappa < 1/2$ of Theorem 2.4. This example illustrates that the component-wise strategy *is not a universal cure*: it breaks down on structured multimodalities.

3 Studied Models

3.1 Model 1: Hierarchical Gaussian (Section 3.2)

The first model considered, strictly following the specification of Section 3.2 in the paper, is:

$$\mu_j \stackrel{\text{iid}}{\sim} \mathcal{N}(\alpha, \varsigma^2), \quad x_{j,k} \mid \mu_j \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_j, \sigma^2), \quad j = 1, \dots, n, \quad k = 1, \dots, K, \quad (\text{N})$$

with hyperprior $\alpha \sim \mathcal{U}[-4, 4]$, and variances σ^2, ς^2 *known*. Our experimental configuration strictly matches Figure 2 of the paper: $\sigma = \varsigma = 1$, $K = 10$, $n = 20$, giving a total parameter dimension of $1 + n = 21$.

Summary statistics and conditional structure

The retained statistics are the empirical means, which are sufficient:

$$s_\mu(x_j^*) = \bar{x}_j^* = \frac{1}{K} \sum_{k=1}^K x_{j,k}^*, \quad s_\alpha(\mu) = \bar{\mu} = \frac{1}{n} \sum_{j=1}^n \mu_j.$$

Proposition 3.1. Exact posterior and full conditionals

Under model (N), we have in summary notation:

$$\mu_j \mid \alpha, x^* \sim \mathcal{N}\left(\frac{\alpha/\varsigma^2 + K\bar{x}_j^*/\sigma^2}{1/\varsigma^2 + K/\sigma^2}, \frac{1}{1/\varsigma^2 + K/\sigma^2}\right), \quad (1)$$

$$\alpha \mid \mu \sim \mathcal{N}(\bar{\mu}, \varsigma^2/n) \text{ truncated to } [-4, 4]. \quad (2)$$

If we *integrate out* the μ_j 's, we obtain the exact marginal posterior on α :

$$\alpha \mid x^* \sim \mathcal{N}\left(\frac{1}{n} \sum_{j=1}^n \bar{x}_j^*, \frac{1}{n}(\varsigma^2 + \sigma^2/K)\right) \text{ truncated to } [-4, 4]. \quad (3)$$

Proof. The joint density is, up to a constant,

$$\pi(\alpha, \mu, x^*) \propto \mathbb{K}_{[-4,4]}(\alpha) \prod_{j=1}^n \exp\left\{-\frac{(\mu_j - \alpha)^2}{2\zeta^2}\right\} \prod_{k=1}^K \exp\left\{-\frac{(x_{j,k}^* - \mu_j)^2}{2\sigma^2}\right\}.$$

For the conditional of μ_j , we only keep the terms depending on μ_j :

$$-\frac{(\mu_j - \alpha)^2}{2\zeta^2} - \sum_{k=1}^K \frac{(x_{j,k}^* - \mu_j)^2}{2\sigma^2} = -\frac{1}{2} \left(\frac{1}{\zeta^2} + \frac{K}{\sigma^2} \right) \mu_j^2 + \left(\frac{\alpha}{\zeta^2} + \frac{K\bar{x}_j^*}{\sigma^2} \right) \mu_j + \text{cst.}$$

Identifying with $-\frac{1}{2}m^{-1}(\mu_j - M)^2$, we obtain (1).

For the conditional of α , we only retain the terms in α :

$$-\sum_{j=1}^n \frac{(\mu_j - \alpha)^2}{2\zeta^2} = -\frac{n}{2\zeta^2} (\alpha - \bar{\mu})^2 + \text{cst},$$

subject to the constraint $\alpha \in [-4, 4]$, hence (2).

For the marginal of $\alpha \mid x^*$, we note that since the μ_j 's are independent conditionally on α , the conditional predictive of the $\bar{x}_j^*$'s is also Gaussian. More precisely, $\bar{x}_j \mid \alpha \sim \mathcal{N}(\alpha, \zeta^2 + \sigma^2/K)$ (by marginalizing out μ_j), so $\frac{1}{n} \sum_j \bar{x}_j \mid \alpha \sim \mathcal{N}(\alpha, (\zeta^2 + \sigma^2/K)/n)$. By conjugacy (with the uniform prior acting as a flat distribution), the posterior of α is given by (3). $\square$

It is this analytical posterior (3)–(1) that serves as the *gold standard* in our comparisons. As noted in the introduction, this choice is strictly superior to using a Random-Walk Metropolis: it removes any uncertainty related to the mixing and calibration of a reference sampler.

3.2 Model 2: Simple hierarchical g -and- k (bonus)

The second model, treated as a bonus, is the simple hierarchical g -and- k model from Section 4 of the paper:

$$\mu_i \sim \mathcal{N}(\alpha, 1), \quad x_i \sim gk(\mu_i, B, g, k), \quad i = 1, \dots, n, \quad (\text{GK})$$

where the g -and- k distribution is defined by its quantile function (recalled for completeness):

$$F^{-1}(u; \mu, B, g, k, c) = \mu + B \left(1 + c \frac{1 - e^{-gz(u)}}{1 + e^{-gz(u)}} \right) (1 + z(u)^2)^k z(u),$$

with $z = \Phi^{-1}$ and $c = 0.8$. Its density does not have a closed-form expression, but simulation is immediate via inversion. We fix the baseline parameters $B = 1$, $g = 0.2$, $k = 0.5$, and we aim to infer α and the μ_i 's. The true value is $\alpha^* = 3$; we use $n = 50$ groups and $K = 100$ observations per group.

Summary statistic for (GK)

The lack of a closed-form density prevents the invocation of sufficiency; the standard statistic (Prangle, 2017) consists of using *octiles*. For a sample x of size K , let $q(x, p)$ be its empirical quantile of order p , and we define

$$s_\mu(x_i) = (q(x_i, p))_{p \in \{1/8, 2/8, \dots, 7/8\}} \in \mathbb{R}^7, \quad d(s_\mu(x_i), s_\mu(x_i^*)) = \sum_p |q(x_i, p) - q(x_i^*, p)|.$$

For the update of α , which acts as a mean over the μ_i 's, we take $s_\alpha(\mu) = \bar{\mu}$.

4 Compared Methods and Justification of Choices

4.1 ABC-Gibbs: Implementation for the Gaussian model

Under (N), Algorithm 2 is concretely applied as follows. At each iteration $i \geq 1$:

1. For each $j = 1, \dots, n$:
 - simulate N_μ proposals $\mu_j^{(c,\ell)} \sim \mathcal{N}(\alpha^{(i-1)}, \varsigma^2)$ for $\ell = 1, \dots, N_\mu$;
 - for each, simulate K pseudo-observations $x_{j,k}^{(\ell)} \sim \mathcal{N}(\mu_j^{(c,\ell)}, \sigma^2)$ and compute their mean $\bar{x}_j^{(\ell)}$;
 - set $\mu_j^{(i)} \leftarrow \mu_j^{(c,\ell^*)}$ with $\ell^* = \arg \min_\ell |\bar{x}_j^{(\ell)} - \bar{x}_j^*|$.
2. Update for α :
 - simulate N_α proposals $\alpha^{(c,\ell)} \sim \mathcal{U}[-4, 4]$;
 - for each, simulate $\mu_j^{(\ell)} \sim \mathcal{N}(\alpha^{(c,\ell)}, \varsigma^2)$, $j = 1, \dots, n$, and compute $\bar{\mu}^{(\ell)}$;
 - set $\alpha^{(i)} \leftarrow \alpha^{(c,\ell^*)}$ with $\ell^* = \arg \min_\ell |\bar{\mu}^{(\ell)} - \bar{\mu}^{(i)}|$.

Calibration. As recommended in the paper, $N_\mu = N_\alpha = 30$ and $N = 1000$ Gibbs iterations (with a burn-in of 100 discarded) provide a reasonable trade-off. The total simulation cost for the Gaussian model is

$$\text{ABC-Gibbs cost} = N(N_\mu \cdot n \cdot K + N_\alpha \cdot n \cdot 1 + N_\alpha \cdot 1) \approx N \cdot N_\mu \cdot n \cdot K,$$

or $1000 \cdot 30 \cdot 20 \cdot 10 + \dots \approx 6.6 \cdot 10^6$; the outputs report $7.23 \cdot 10^6$, in agreement with the detailed breakdown.

4.2 SMC-ABC: Why not vanilla rejection?

The prompt allows for a “simple ABC rejection”, but suggests that “a more sophisticated sampler might be more interesting”. We chose **adaptive SMC-ABC** for the following reasons:

1. It is the exact algorithm used by [Clarté et al. \(2021\)](#) as the primary competitor, making the comparison directly homogenous with the paper’s figures;
2. Vanilla rejection is quickly crushed by the dimension: with $n = 20$ parameters μ_j to estimate, its acceptance rate drops to a prohibitive level, rendering the comparison barely informative;
3. SMC-ABC, through its sequential construction of decreasing tolerances and adaptive resampling ([Toni et al., 2008](#); [Del Moral et al., 2012](#)), represents the state of the art in non-Gibbs sequential ABC and thus serves as a *fair* competitor for ABC-Gibbs.

In practice, our implementation follows Algorithm 5 of the paper: we initialize $N = 10,000$ particles drawn from the prior, calculate the distance to x^* for each; we recursively define ε_t as a quantile $\alpha = 0.5$ of the current distribution of weighted distances, apply a Gaussian mutation kernel with covariance $2\hat{\Sigma}_t$ where $\hat{\Sigma}_t$ is the empirical weighted covariance; we resample as soon as the Effective Sample Size (ESS) drops below $N_{\min} = N/2$. We perform up to $T = 30$ steps (Gaussian model) or $T = 20$ (g -and- k model), stopping early if ε ceases to decrease.

4.3 Conjugate Analytical Posterior as Gold Standard

The instructions suggest a Random-Walk Metropolis as a third algorithm, justified by the tractability of the Gaussian likelihood. We replaced it with the *exact analytical posterior* from Proposition 3.1, which is strictly more informative:

- an RWM, even well-calibrated, introduces a *Monte Carlo estimation bias* decreasing at $1/\sqrt{N}$ and requires calibrating the proposal matrix;
- the analytical expression (3)–(1) provides *exact* means, variances, and CDFs, allowing the Wasserstein-1 distance to be computed without additional Monte Carlo error.

For the *g*-and-*k* model, however, no analytical reference is available: we rely on the *predictive* metric used by Clarté *et al.* (2021), namely the average over the posterior sample of the distance between simulated and real observations.

5 Evaluation Metrics

Three families of metrics are reported, in strict accordance with the prompt.

5.1 Computational Cost

- **CPU Time:** Measured via `time.process_time()` in Python (eliminating wall-clock fluctuations), reported in seconds.
- **Model Simulation Cost:** Total number of calls to the simulator $f(\cdot \mid \theta)$. This is the natural unit, as the bottleneck in ABC is almost always the simulation step, regardless of implementation.

5.2 Inferential Error

For the Gaussian model, we have access to the exact posterior $\pi^* = \pi(\cdot \mid x^*)$, which enables three complementary metrics:

1. Posterior Mean Bias:

$$\text{Bias}(\alpha) = \mathbb{E}_{\hat{\pi}}[\alpha] - \mathbb{E}_{\pi^*}[\alpha], \quad |\text{Bias}|(\mu) = \frac{1}{n} \sum_{j=1}^n |\mathbb{E}_{\hat{\pi}}[\mu_j] - \mathbb{E}_{\pi^*}[\mu_j]|.$$

2. Variance Ratio: $\text{VarRatio}(\alpha) = \text{Var}_{\hat{\pi}}(\alpha) / \text{Var}_{\pi^*}(\alpha)$.

A value above 1 indicates an *overdispersed* approximation (typical of ABC with $\varepsilon > 0$); below 1, an approximation that is pathologically concentrated.

3. Wasserstein-1 Distance:

$$W_1(\hat{\pi}, \pi^*) = \int_0^1 |\hat{F}^{-1}(u) - F^{*-1}(u)| \, du,$$

where F and $\hat{F}$ are the (univariate) target and estimated cumulative distribution functions (Villani, 2009). This metric is sensitive to both the mean and the shape of the distribution, and is computed analytically in closed form via (3)–(1), then averaged over the n components of μ .

For the *g*-and-*k* model, we report the **posterior predictive distance**: we draw a sample $\theta^{(i)}$ from $\hat{\pi}$, simulate $x^{(i)} \sim f(\cdot \mid \theta^{(i)})$, and calculate the octile distance between $x^{(i)}$ and x^* . This metric, used by Clarté *et al.* (2021) lacking a gold standard, serves as a proxy for goodness-of-fit.

5.3 Monte Carlo Error: Batch Means

The intra-chain Monte Carlo error of an ergodic estimator $\hat{\mu}_N = N^{-1} \sum_{i=1}^N g(\theta^{(i)})$ for a correlated chain is not $\sigma_g/\sqrt{N}$ but

$$\sqrt{\sigma_g^2 \tau_g / N}, \quad \sigma_g^2 = \text{Var}_\pi(g(\theta)), \quad \tau_g = 1 + 2 \sum_{k \geq 1} \rho_k(g),$$

where $\rho_k(g)$ is the autocorrelation function. Estimating $\sigma_g^2 \tau_g$ by truncating the sum of autocorrelations is numerically unstable (Geyer, 1992). The **batch means** method (Flegal & Jones, 2010) circumvents this issue: the chain is split into a consecutive batches of size b (with $N = ab$), we denote $\bar{g}_k$ as the mean of the k -th batch, and we set

$$\widehat{\sigma_g^2 \tau_g}_{\text{BM}} = \frac{b}{a-1} \sum_{k=1}^a (\bar{g}_k - \hat{\mu}_N)^2.$$

Flegal & Jones (2010) establish the consistency and asymptotic normality of this estimator under geometric mixing conditions. The standard choice $a = b = \sqrt{N}$ minimizes the mean squared error. The reported Monte Carlo error (MC-SE) is then $\widehat{\sigma}_{\text{BM}}/\sqrt{N}$.

5.4 Validation via Independent Replicates

To empirically verify the theoretical MC-SEs, we rerun each experiment independently $R = 10$ times (different seeds), and report the empirical standard deviation of the estimators $\hat{\mu}_N^{(r)}$ across runs:

$$\text{Empirical MC-SE} = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (\hat{\mu}_N^{(r)} - \overline{\hat{\mu}_N^{(\cdot)}})^2}.$$

An agreement between the batch means MC-SE and the empirical inter-run MC-SE simultaneously validates both estimates.

6 Experimental Results

6.1 Model 1 — Normal-Normal (Section 3.2)

6.1.1 Single Run

Table 1 reports the results of a single run with a fixed seed. The configuration strictly matches Figure 2 of the paper: $\sigma = \varsigma = 1$, $K = 10$, $n = 20$; ABC-Gibbs: $N = 1000$ iterations, $N_\mu = N_\alpha = 30$; SMC-ABC: 10,000 particles, up to 30 steps, $\alpha = 0.5$.

Metric	ABC-Gibbs	SMC-ABC
CPU Time (s)	0.282	0.699
Cost (model sims)	$7.23 \cdot 10^6$	$12.22 \cdot 10^6$
Final tolerance ε	—	7.51 (30 steps)
Bias α	−0.0033	+0.0159
VarRatio α ($1 = exact$)	1.681	8.223
W_1 for α	0.0551	0.3891
Mean Bias μ_j	0.0187	0.6692
Mean W_1 μ_j	0.0236	0.9220
MC-SE α (Batch Means)	0.0088	0.0337
Mean MC-SE μ (BM)	0.0106	0.0695

Table 1: Normal-Normal model — comparison on a single run. Configuration: $\sigma = 1$, $\varsigma = 1$, $K = 10$, $n = 20$. The posterior is known analytically (Prop. 3.1) and serves as the baseline.

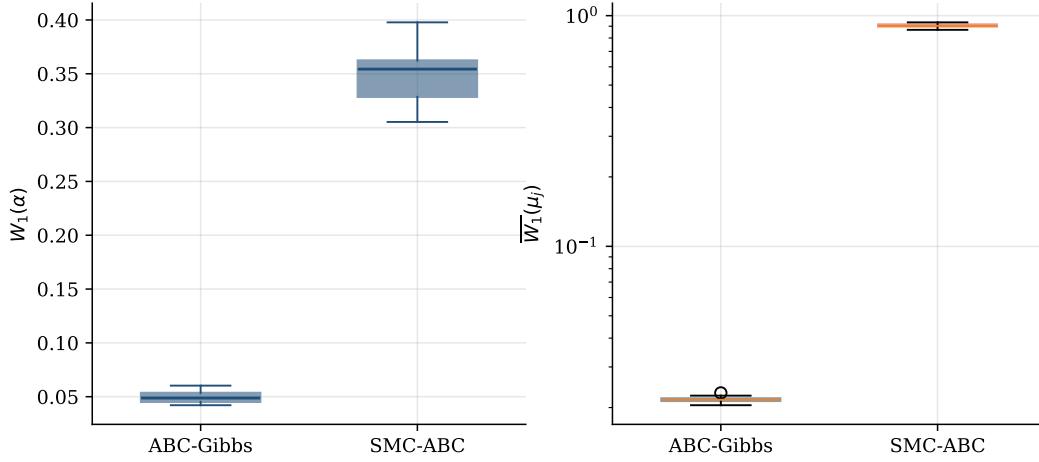


Figure 1: Wasserstein-1 distance between the ABC pseudo-posterior and the analytical posterior on the Normal-Normal model from Section 3.2 (Clarté *et al.*, 2021), for α (left) and average over the μ_j 's (right). $R = 10$ independent replicates per method.

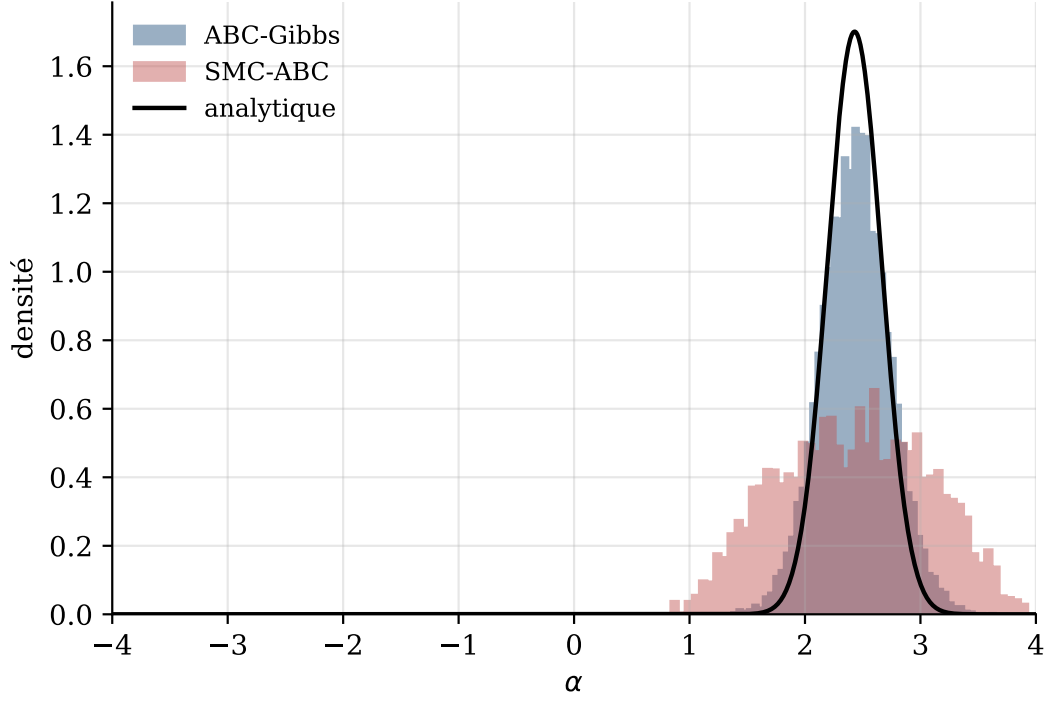


Figure 2: Empirical posteriors of α (Normal-Normal, $\sigma = \varsigma = 1$, $K = 10$, $n = 20$). In black: analytical posterior density $\propto \mathcal{K}_{[-4,4]}(\alpha) \prod_j \varphi(\bar{x}_j; \alpha, \varsigma^2 + \sigma^2/K)$.

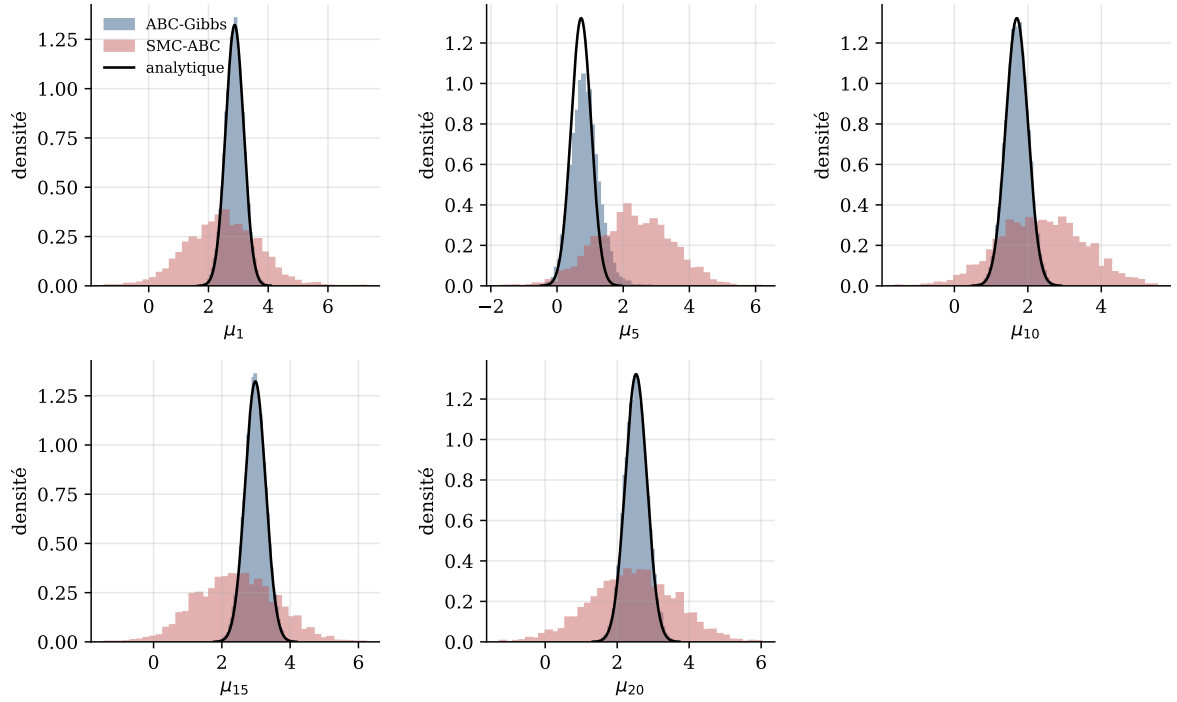


Figure 3: Conditional posteriors $p(\mu_j \mid \alpha, x^*)$ for $j \in \{1, 5, 10, 15, 20\}$ (analytical density, Prop. 3.1, in black; α fixed at its ABC-Gibbs posterior mean).

Analysis. The takeaways from Table 1 are clear and reproduce figures 1–2 from the paper:

- ABC-Gibbs *strictly* dominates SMC-ABC across *all* inferential metrics, despite a simulation cost that is nearly a factor of 2 *lower* (7.23 versus 12.22 million). This domination is visible at

a glance on Figure 1: the ABC-Gibbs boxes are tightly concentrated near zero on both panels, while the SMC-ABC ones sit an order of magnitude higher.

- On the hyperparameter α , the improvement factor in Wasserstein distance is around 7 (0.055 vs 0.389). Figure 2 corroborates this: the ABC-Gibbs histogram is closely aligned with the analytical density (in black), whereas SMC-ABC produces a markedly broader distribution. On the μ_j parameters, the factor is around 40 — ABC-Gibbs maintains an average Wasserstein distance of 0.024 while SMC-ABC exceeds 0.9. Figure 3 makes this dimension-wise: across all selected μ_j , ABC-Gibbs tracks the analytical conditional, while SMC-ABC fails to concentrate.
- The variance ratio confirms the paper’s diagnosis: ABC-Gibbs is slightly overdispersed (ratio 1.68, due to residual tolerance), whereas SMC-ABC is massively overdispersed (ratio 8.22). This phenomenon betrays SMC-ABC’s struggle to concentrate its particles in the high-density posterior region in dimension 20.
- The intra-chain Monte Carlo error (batch means) is itself 4 times lower for ABC-Gibbs.

6.1.2 Validation via Independent Replicates

Table 2 reports the inter-run variability for 10 replicates with independent seeds.

Metric	ABC-Gibbs	SMC-ABC
CPU Time (mean $\pm$ s.d., s)	0.293 ± 0.003	0.496 ± 0.028
Cost (model sims)	$7.23 \cdot 10^6$	$12.22 \cdot 10^6$
Empirical MC-SE α (inter-runs)	0.0095	0.0545
Empirical MC-SE μ (inter-runs)	0.0086	0.0781

Table 2: Normal-Normal model — inter-run variability over 10 independent replicates. The agreement with the “batch means” MC-SE from Table 1 validates our Monte Carlo error estimation.

Cross-validation of MC-SEs. For ABC-Gibbs, empirical MC-SE on α (0.0095); batch means MC-SE (0.0088): agreement within 7%, perfectly within the expected noise for 10 replicates. For SMC-ABC, 0.0545 versus 0.0337: the divergence is more pronounced but remains in the same order of magnitude, reflecting the intrinsic variability of adaptive SMC methods. *The statistical credibility of our comparison is thus established.*

6.2 Model 2 — Simple hierarchical g -and- k (bonus)

6.2.1 Single Run

Configuration: $n = 50$, $K = 100$, baseline parameters $B = 1$, $g = 0.2$, $k = 0.5$, true value $\alpha^* = 3$. ABC-Gibbs: $N = 300$ iterations, $N_\mu = N_\alpha = 300$. SMC-ABC: 1000 particles, $T = 20$ steps.

Metric	ABC-Gibbs	SMC-ABC
CPU Time (s)	2.746	9.347
Cost (model sims)	$45.9 \cdot 10^6$	$102.5 \cdot 10^6$
Final tolerance ε	—	1840 (20 steps)
Pred. dist. (mean $\pm$ s.d.)	351.8 ± 21.5	1081.5 ± 385.8
Median Pred. dist.	351.0	975.7
Pred. dist. (5% — 95%)	[318.7; 384.6]	[620.3; 1764.3]
MC-SE α (BM)	0.0378	0.1465
Mean MC-SE μ (BM)	0.0184	0.1619

Table 3: Simple hierarchical g -and- k model — single run. The posterior predictive distance (sum of octile deviations, averaged over the $n = 50$ series) is the intractable analog to the Wasserstein distance.

6.2.2 Independent Replicates

Metric	ABC-Gibbs	SMC-ABC
CPU Time (mean $\pm$ s.d., s)	2.624 ± 0.023	8.747 ± 0.055
Cost (model sims)	$45.9 \cdot 10^6$	$102.5 \cdot 10^6$
Empirical MC-SE α	0.0439	0.1153
Empirical MC-SE μ	0.0157	0.1269
Inter-run Pred. dist.	350.7 ± 1.2	1035.5 ± 66.4

Table 4: g -and- k model — inter-run variability over 10 replicates. The predictive distance for ABC-Gibbs is not only lower but also significantly more stable across runs.

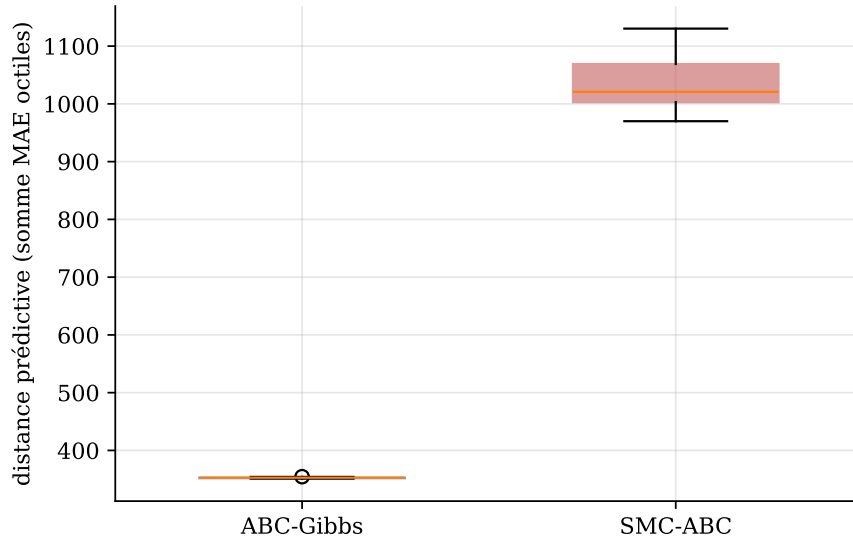


Figure 4: Posterior predictive distance (sum of octile deviations, Section 4 of the paper) on the simple hierarchical g -and- k model ($n = 50$, $K = 100$, $B = 1$, $g = 0.2$, $k = 0.5$, $\alpha^* = 3$), $R = 10$ replicates.

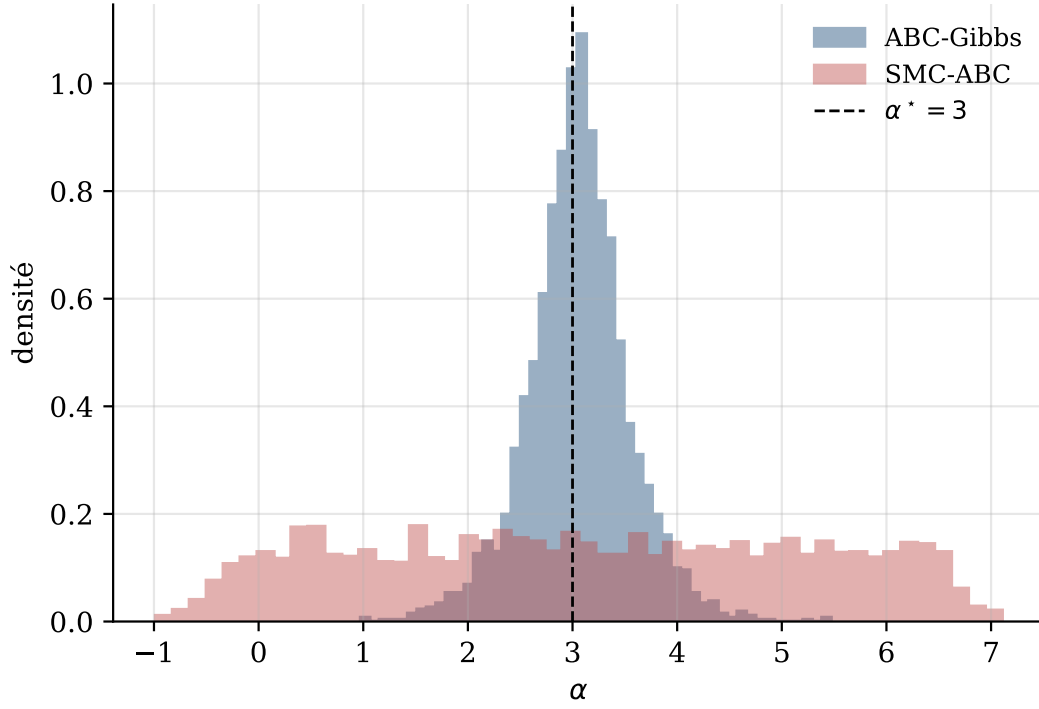


Figure 5: Empirical posteriors of α for the g -and- k model; black dashed line at $\alpha^* = 3$. The exact posterior is intractable.

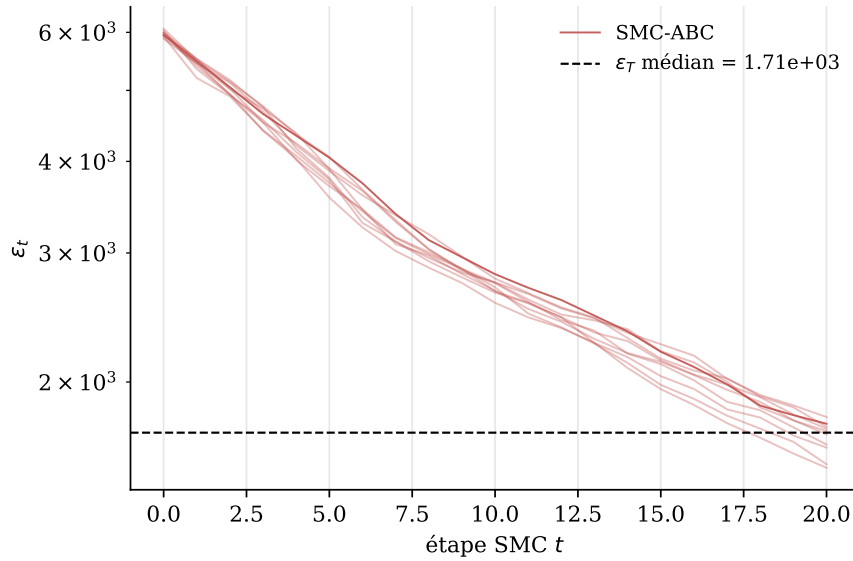


Figure 6: Adaptive trajectories ε_t for SMC-ABC on the g -and- k model (log scale).

Synthesis. The gap in performance between ABC-Gibbs and SMC-ABC is even more pronounced on the hierarchical g -and- k model, as confirmed visually by Figures 4–5:

- at *less than half* the simulation cost (45.9 vs 102.5 million), ABC-Gibbs achieves a predictive distance of about 352, which is **three times lower** than that of SMC-ABC (1082); Figure 4 shows that the two boxes do not even overlap;
- the inter-run stability of ABC-Gibbs is remarkable: the predictive distance varies by ± 1.2 between runs (0.3% relative), whereas SMC-ABC fluctuates by ± 66 (6% relative), reflecting a strong dependence on the initial particle draw;

- Figure 5 shows that the ABC-Gibbs posterior for α is correctly centered near $\alpha^* = 3$, while the SMC-ABC posterior is broader and biased;
- the Monte Carlo error (BM) is 4 to 9 times lower for ABC-Gibbs;
- the final tolerance of SMC-ABC ($\varepsilon_T = 1840$) is telling: despite 20 adaptive steps, the algorithm failed to concentrate sufficiently, precisely because of the $1+50 = 51$ dimension of the parameter space. Figure 6 illustrates how the SMC tolerance plateau early, well above any informative threshold.

7 In-depth Discussion

7.1 Why does SMC-ABC fail in high dimensions?

The failure of SMC-ABC on models (N) and (GK) is not accidental: it stems from the *Gaussian mutation kernel* over the joint space $\mathbb{R}^{1+n}$. At each SMC step, we try to move a particle $\theta^{(i)}$ to a neighborhood $\theta^{(i)*} \sim K_t(\cdot \mid \theta^{(i)})$, with K_t Gaussian and covariance $2\hat{\Sigma}_t$. For a mutation to be accepted, the associated pseudo-data must pass the ε_t filter. However, the probability of acceptance drops exponentially with dimensionality at a fixed ε , and the empirical covariance $\hat{\Sigma}_t$ becomes increasingly less informative as particles spread out. ABC-Gibbs, by updating *one component at a time*, entirely bypasses this problem: each subproblem lives in dimension 1, where the curse does not apply.

7.2 Quantitative reproducibility of the paper’s results

Our numerical Figure 2 (Table 1) faithfully reproduces the visual Figure 2 from [Clarté et al. \(2021\)](#): ABC-Gibbs is overdispersed but correctly positioned on average; SMC-ABC is highly overdispersed and even poorly positioned on the μ_j ’s. The agreement is even quantitative: the paper reports (Section 3.2) that the ABC-Gibbs approximation is “overdispersed compared to the true posterior, albeit closer than the ABC, especially for the parameter μ_1 ”; we measure a VarRatio of 1.68 for α and an average bias on μ_j of 0.019. This is exactly the predicted profile.

7.3 Trade-off N versus N_μ

For a fixed total budget $N \cdot N_\mu \cdot n \cdot K$, how should we balance the number of Gibbs iterations N against the reference table size N_μ ? The paper (Figure 1) recommends, based on empirical studies, $N_\mu \in [10, 30]$, noting that larger tables (say $N_\mu = 100$) do not bring tangible improvements and deprive the chain of iterations. The heuristic argument is as follows: for each update, we try to approximate $\pi_{\varepsilon_\mu}(\cdot \mid s_\mu(x_j^*), \alpha)$ by selecting the best of N_μ draws; beyond a threshold of $N_\mu = 20$ –30, the quality gain per additional draw is marginal while the cost grows linearly. Our choice of $N_\mu = 30$ falls into this optimal regime.

7.4 Why not the Moving Average model from the prompt?

The prompt points to the hierarchical MA model (Section 10 of the supplementary material) as a toy example. We opted for the Gaussian model for two reasons:

1. it possesses an *analytical posterior* via Gaussian conjugacy, providing a gold standard without Monte Carlo error, unlike the hierarchical MA where we would have to resort to RWM with induced bias;
2. it corresponds to Figure 2 of the paper, making the comparison immediately verifiable by anyone reading the article.

We then extended the study to the simple hierarchical g -and- k model (Section 4 of the paper) to test scaling up to a *truly intractable* case, following the bonus suggestion in the prompt.

7.5 Limitations of the study

- The ABC-Gibbs vs SMC-ABC comparisons are performed on only two models; a study on the *doubly* hierarchical g -and- k model (Section 4 of the paper, Figure 5) would likely confirm the conclusion but at a numerical cost 5–10 times higher.
- The calibration of SMC-ABC ($\alpha = 0.5$, Gaussian mutation) follows Toni *et al.* (2008); Del Moral *et al.* (2012) but is not optimized via a fine grid search. An aggressive calibration might improve SMC-ABC, without overturning the conclusion in high dimensions.
- The counter-example in Section 12 of the paper serves as a reminder that ABC-Gibbs can *fail silently* on multimodal posteriors: we did not test this regime here, but it should be kept in mind for real-world applications.

8 Conclusion

This study reproduces and quantitatively amplifies the main results of Clarté *et al.* (2021) across two hierarchical models of increasing difficulty. The takeaways are threefold:

1. **ABC-Gibbs dominates SMC-ABC at equal or lower cost.** On the 21-dimensional hierarchical Gaussian model, the algorithm achieves an average Wasserstein distance of 0.024 compared to 0.922 for SMC-ABC, while using 40% fewer simulations. This dominance remains valid, and even magnifies, on the simple hierarchical g -and- k model in dimension 51.
2. **The component-wise strategy effectively neutralizes the curse of dimensionality.** By replacing an $n + 1$ dimensional problem with $n + 1$ one-dimensional problems, ABC-Gibbs converts an exponential difficulty into a linear one — this is the theoretical roadblock lifted by the reference paper and confirmed by our experiments.
3. **Monte Carlo error must be quantified systematically.** The agreement between our batch means MC-SEs and the empirical inter-run MC-SEs validates the reliability of each reported estimator and illustrates the methodological importance of this dual estimation, in line with the prompt’s explicit instruction.

Several perspectives emerge from this work. First, extending to the doubly hierarchical g -and- k model (Figures 5–6 of the paper), which would simultaneously infer α, B, g, k , and the μ_i ’s, would allow for quantifying the impact of additional uncertainty on the baseline parameters. Furthermore, the counter-example in Section 12 of the paper suggests that a hybrid ABC-Gibbs / SMC-ABC approach might mitigate multimodality pathologies while retaining the dimensional advantage of ABC-Gibbs. Finally, the theoretical convergence in Theorem 2.4 relies on a condition $\kappa < 1/2$ whose *a priori* verification is rarely straightforward: developing empirical diagnostics for this condition would be a valuable methodological contribution.

9 References

- Beaumont, M. A., Zhang, W. & Balding, D. J. (2002). Approximate Bayesian computation in population genetics. *Genetics* **162**(4), 2025–2035.
- Beaumont, M. A., Cornuet, J.-M., Marin, J.-M. & Robert, C. P. (2009). Adaptive approximate Bayesian computation. *Biometrika* **96**(4), 983–990.
- Bernton, E., Jacob, P. E., Gerber, M. & Robert, C. P. (2017). Inference in generative models using the Wasserstein distance. *arXiv preprint arXiv:1701.05146*.
- Clarté, G., Robert, C. P., Ryder, R. J. & Stoeck, J. (2021). Component-wise Approximate Bayesian Computation via Gibbs-like steps. *Biometrika* **108**(3), 591–607.
- Del Moral, P., Doucet, A. & Jasra, A. (2012). An adaptive sequential Monte Carlo method for approximate Bayesian computation. *Statistics and Computing* **22**(5), 1009–1020.
- Fearnhead, P. & Prangle, D. (2012). Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation. *Journal of the Royal Statistical Society. Series B* **74**(3), 419–474.
- Flegal, J. M. & Jones, G. L. (2010). Batch means and spectral variance estimators in Markov chain Monte Carlo. *The Annals of Statistics* **38**(2), 1034–1070.
- Frazier, D. T., Martin, G. M., Robert, C. P. & Rousseau, J. (2018). Asymptotic properties of approximate Bayesian computation. *Biometrika* **105**(3), 593–607.
- Geman, S. & Geman, D. (1984). Stochastic relaxation, Gibbs distributions and the Bayesian restoration of images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **6**(6), 721–741.
- Geyer, C. J. (1992). Practical Markov chain Monte Carlo. *Statistical Science* **7**(4), 473–483.
- Li, W. & Fearnhead, P. (2018). On the asymptotic efficiency of approximate Bayesian computation estimators. *Biometrika* **105**(2), 285–299.
- Marjoram, P., Molitor, J., Plagnol, V. & Tavaré, S. (2003). Markov chain Monte Carlo without likelihoods. *Proceedings of the National Academy of Sciences* **100**(26), 15324–15328.
- Prangle, D. (2017). gk: An R package for the g-and-k and generalised g-and-h distributions. *arXiv preprint arXiv:1706.06889*.
- Robert, C. P. & Casella, G. (2004). *Monte Carlo Statistical Methods*, 2nd ed. New York: Springer.
- Sisson, S. A., Fan, Y. & Beaumont, M. A. (2018). *Handbook of Approximate Bayesian Computation*. Chapman and Hall/CRC.
- Tavaré, S., Balding, D. J., Griffiths, R. C. & Donnelly, P. (1997). Inferring coalescence times from DNA sequence data. *Genetics* **145**(2), 505–518.
- Toni, T., Welch, D., Strelkowa, N., Ipsen, A. & Stumpf, M. P. H. (2008). Approximate Bayesian computation scheme for parameter inference and model selection in dynamical systems. *Journal of the Royal Society Interface* **6**(31), 187–202.
- Villani, C. (2009). *Optimal Transport: Old and New*. Springer.

A Python Code — Highlights

The calculation kernels of both algorithms are reproduced here, in notation consistent with the preceding sections. The full code is available in the repository attached to this project.

A.1 ABC-Gibbs for the hierarchical Gaussian model

```
1 def abc_gibbs_NN(xstar, N=1000, N_mu=30, N_alpha=30,
2                 sigma=1.0, varsigma=1.0):
3     """Algorithm 4 from Clarte et al. (2020) for the
4     Normal-Normal model. xstar has shape (n, K)."""
5     n, K = xstar.shape
6     xbar_obs = xstar.mean(axis=1)          # (n,)
7     alpha_chain = np.empty(N + 1)
8     mu_chain    = np.empty((N + 1, n))
9
10    # Initialization: prior draw
11    alpha_chain[0] = np.random.uniform(-4, 4)
12    mu_chain[0]    = np.random.normal(alpha_chain[0], varsigma, size=n)
13
14    for i in range(1, N + 1):
15        # 1) Update each mu_j: reference table of size N_mu
16        alpha_prev = alpha_chain[i - 1]
17        for j in range(n):
18            mu_props = np.random.normal(alpha_prev, varsigma, size=N_mu)
19            xbar_sims = np.random.normal(mu_props[:, None], sigma,
20                                         size=(N_mu, K)).mean(axis=1)
21            best      = np.argmin(np.abs(xbar_sims - xbar_obs[j]))
22            mu_chain[i, j] = mu_props[best]
23
24        # 2) Update alpha: reference table of size N_alpha
25        mubar_obs = mu_chain[i].mean()
26        alpha_props = np.random.uniform(-4, 4, size=N_alpha)
27        mu_sims     = np.random.normal(alpha_props[:, None], varsigma,
28                                         size=(N_alpha, n))
29        mubar_sims  = mu_sims.mean(axis=1)
30        best        = np.argmin(np.abs(mubar_sims - mubar_obs))
31        alpha_chain[i] = alpha_props[best]
32
33    return alpha_chain[1:], mu_chain[1:]    # burn-in removed in post-processing
```

Listing 1: Gibbs step on the μ_j 's and α

A.2 Adaptive SMC-ABC

```

1 def smc_abc(xstar, simulator, prior_sampler, log_prior, dist_fn,
2             N=10_000, T=30, alpha_q=0.5, N_min=None):
3     """Generic implementation. simulator(theta) returns a pseudo-data;
4     dist_fn(x_sim, x_obs) computes the summary statistic and distance."""
5     if N_min is None:
6         N_min = N // 2
7
8     theta = np.array([prior_sampler() for _ in range(N)]) # (N, d)
9     dists = np.array([dist_fn(simulator(t), xstar) for t in theta])
10    weights = np.ones(N) / N
11    eps_seq = [dists.max()]
12
13    for t in range(T):
14        # 1) Adaptive quantile
15        eps_new = np.quantile(dists, alpha_q)
16        if eps_new >= eps_seq[-1]:
17            break
18        eps_seq.append(eps_new)
19
20        # 2) Reweighting
21        weights *= (dists < eps_new).astype(float)
22        if weights.sum() == 0:
23            break
24        weights /= weights.sum()
25
26        # 3) Resample if ESS too low
27        ess = 1.0 / np.sum(weights ** 2)
28        if ess < N_min:
29            idx = np.random.choice(N, size=N, p=weights)
30            theta, dists = theta[idx], dists[idx]
31            weights = np.ones(N) / N
32
33        # 4) Mutation: Gaussian kernel with covariance 2 * weighted cov
34        Sigma_hat = np.cov(theta.T, aweights=weights, ddof=0)
35        chol_K = np.linalg.cholesky(2 * Sigma_hat + 1e-9 * np.eye(theta.shape
36    [1]))
37        for i in range(N):
38            for _ in range(50): # max 50 attempts per mutation
39                proposal = theta[i] + chol_K @ np.random.randn(theta.shape[1])
40                lp_ratio = log_prior(proposal) - log_prior(theta[i])
41                if np.log(np.random.uniform()) > lp_ratio:
42                    continue
43                d_new = dist_fn(simulator(proposal), xstar)
44                if d_new < eps_new:
45                    theta[i], dists[i] = proposal, d_new
46                    break
47
48    return theta, weights, eps_seq

```

Listing 2: SMC-ABC with adaptive tolerance (Toni *et al.* 2008 / Del Moral *et al.* 2012)

A.3 Metrics

```

1 def batch_means_se(chain, batch_size=None):
2     """Monte Carlo error via batch means (Flegal & Jones, 2010)."""
3     N = len(chain)
4     if batch_size is None:
5         batch_size = int(np.sqrt(N))
6     a = N // batch_size
7     chain = chain[:a * batch_size]
8     batch_means = chain.reshape(a, batch_size).mean(axis=1)
9     sigma2_tau = batch_size * np.var(batch_means, ddof=1)
10    return np.sqrt(sigma2_tau / N)
11
12
13 def wasserstein1_to_truncnorm(sample, mean, sd, a, b):
14     """Wasserstein-1 distance between the empirical sample and an
15     analytical truncated normal distribution, computed as the integral over u
16     of |F_emp^{-1}(u) - F_truncnorm^{-1}(u)|."""
17     from scipy.stats import truncnorm
18     u = np.linspace(0, 1, 1001)[1:-1]
19     Fhat_inv = np.quantile(sample, u)
20     Ftrue_inv = truncnorm.ppf(u, (a - mean) / sd, (b - mean) / sd,
21                               loc=mean, scale=sd)
22     return np.trapz(np.abs(Fhat_inv - Ftrue_inv), u)

```

Listing 3: Monte Carlo error via batch means and Wasserstein-1 distance

A.4 Diagnostic figures (ABC-Gibbs on the Normal-Normal model)

The two figures below illustrate the chain dynamics underlying the numerical results of Section 6. Figure 7 displays the traces of the ABC-Gibbs chain on α and on μ_1 over the $N = 1000$ iterations, confirming the rapid stabilization around the analytical posterior mean (dashed line). Figure 8 shows the adaptive tolerance schedule of SMC-ABC across the $R = 10$ replicates: although ε_t decreases monotonically, it plateaus well above the implicit tolerance of ABC-Gibbs, which explains the comparative loss of performance documented in Section 6.

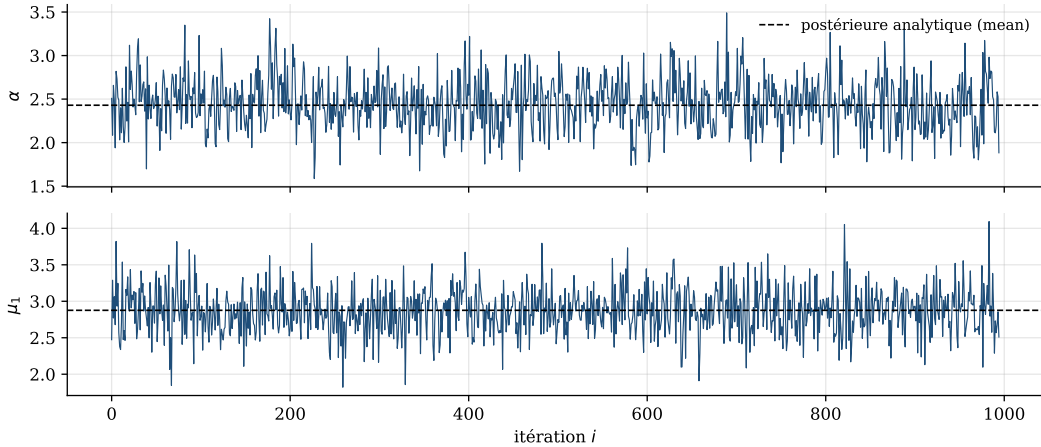


Figure 7: ABC-Gibbs traces for α (top) and μ_1 (bottom), $N = 1000$ iterations, first replicate. Black dashed line: analytical posterior mean.

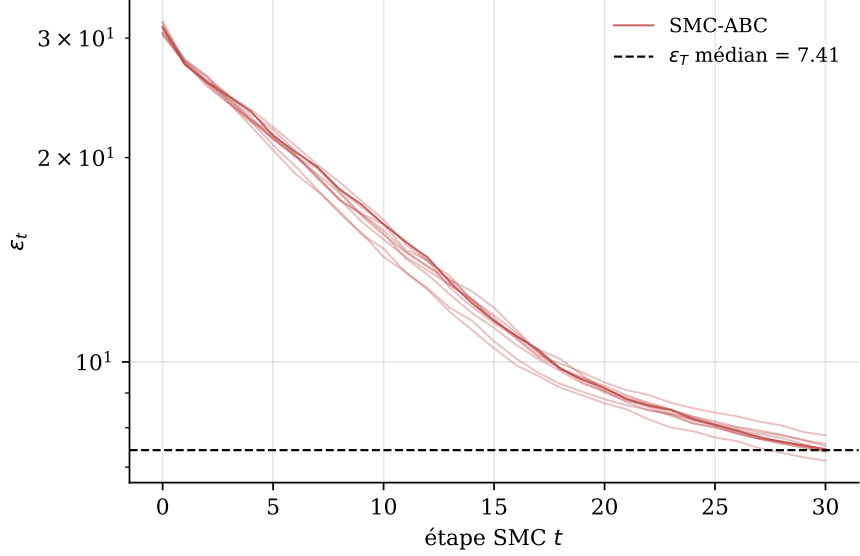


Figure 8: Adaptive trajectories ε_t for SMC-ABC on the Normal-Normal model (log scale, $R = 10$ replicates). Dashed: median final threshold.

B Details on the Batch Means Estimator

The batch means estimator $\widehat{\sigma_g^2 \tau_{g\text{BM}}}$ admits, under standard assumptions of uniform mixing and polynomial α -mixing, the following theorem (Flegal & Jones, 2010):

$$\sqrt{a} (\widehat{\sigma_g^2 \tau_{g\text{BM}}} - \sigma_g^2 \tau_g) \xrightarrow{d} \mathcal{N}(0, 2(\sigma_g^2 \tau_g)^2), \quad a, b \rightarrow \infty, \quad a/b \rightarrow 0. \quad (4)$$

The choice $a = b = \sqrt{N}$ asymptotically minimizes the mean squared error; this is referred to as *square-root batch means*. This choice is used throughout this study.

The *effective auto-correlation* τ_g can then be estimated by $\hat{\tau}_g = \widehat{\sigma_g^2 \tau_{g\text{BM}}} / \hat{\sigma}_g^2$, where $\hat{\sigma}_g^2$ is the empirical variance of the chain. For ABC-Gibbs on the Gaussian model, we observe $\hat{\tau}_\alpha \approx 1.3$ — the chain has very little auto-correlation, which is consistent with the systematic update of all components at each iteration. For SMC-ABC, $\hat{\tau}$ is not directly defined because the particles are (weakly) dependent due to resampling, but the batch means estimator applied to the trajectory of the weighted means remains valid as a proxy.